

number of executions and the time your code takes to run (e.g., measured in milliseconds). There's no need to worry about idle server time.

Event-driven architecture:

Serverless applications are often event driven. Your functions are triggered by specific events, such as:

- API gateway requests (e.g., someone accessing your web API)
- Changes in databases (e.g., a new user added)
- Messages in queues (e.g., an order being placed)

The serverless platform listens for these events and automatically triggers the relevant functions.

Abstracted infrastructure management:

In serverless computing, the cloud provider (e.g., AWS Lambda, Google Cloud Functions) manages the underlying infrastructure. You don't need to worry about server provisioning, patching, scaling, or even monitoring. The cloud provider handles this, ensuring high availability and fault tolerance.

Stateless nature: Serverless functions are stateless by design. This means each time a function is triggered, it doesn't retain any information from previous executions. If your application needs to keep track of state or session data, you store it externally, typically in a database or other stateful services.

Rapid development and deployment: Serverless allows developers to focus purely on writing code, without the hassle of managing servers. This accelerates the development process because you can deploy small, independent functions quickly and scale them as needed.

Changing the cloud landscape

Serverless computing is changing the cloud landscape by providing a more efficient, scalable, and developer-friendly approach to application development and deployment. Here's how.

Simplified development and operations: Traditional cloud models require developers to manage

infrastructure, including provisioning servers, setting up storage, and scaling resources based on demand. With serverless computing, this complexity is abstracted away. Developers can now write and deploy code without managing the underlying infrastructure. Cloud providers handle provisioning, scaling, patching, and even fault tolerance.

This simplification means:

- Faster time-to-market for applications.
- Reduced operational overhead, allowing developers to focus on writing business logic rather than managing infrastructure.

Scalability and flexibility: One of the most impactful changes brought by serverless computing is its ability to scale automatically. Serverless platforms like AWS Lambda, Azure Functions, and Google Cloud Functions can automatically scale applications in response to demand, from zero to millions of requests, without manual intervention. These platforms

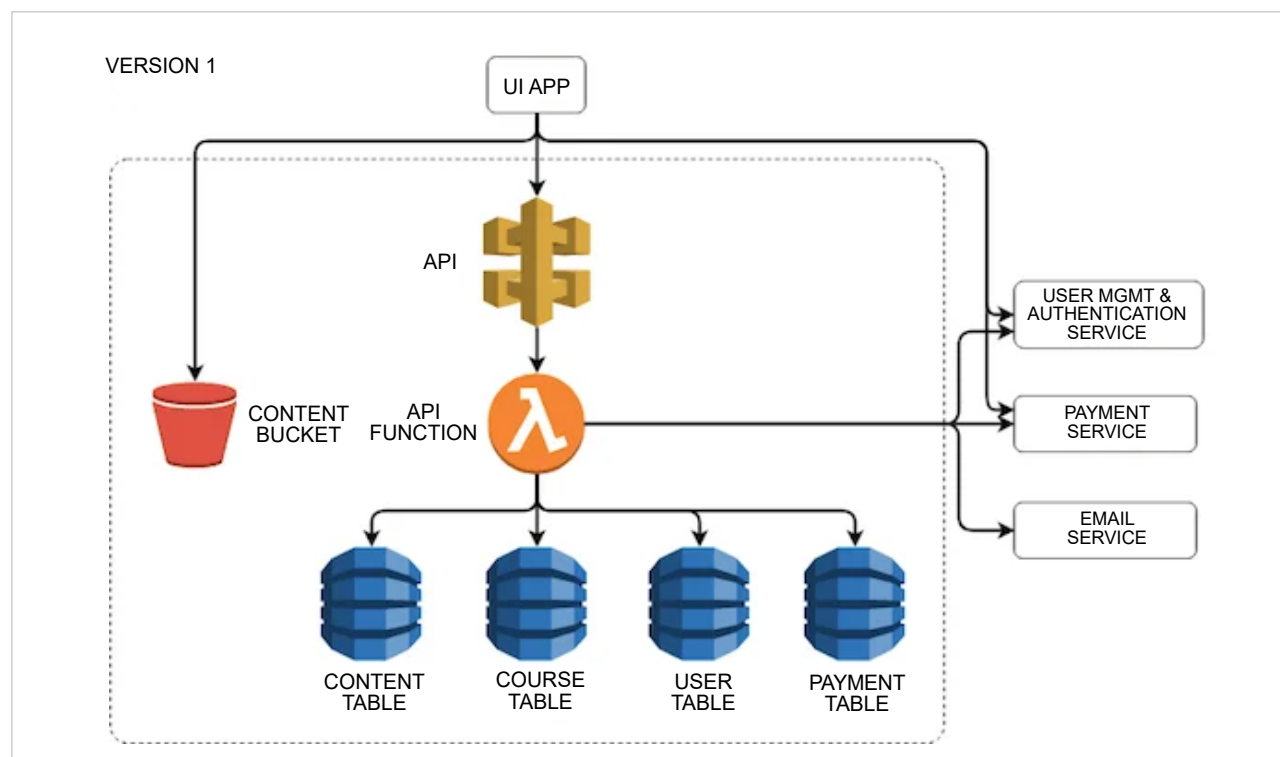


Figure 5: Abstracted infrastructure management